1.2 Cardio-respiratory system and cardiovascular systems	1.2.1	Knowledge, understanding and application of the anatomy and physiology of the cardiovascular, circulatory and respiratory systems in physical activity. Understanding of how they function individually and in conjunction with each other.
	1.2.2	The structure and function of the respiratory system to include the larynx, pharynx, trachea, bronchus, bronchiole, alveoli.
	1.2.3	The physiology of the respiratory system as a mechanical process of ventilation (inspiration and expiration). The cause and effect process, including the role of pressure gradients, partial pressure (pp) and diffusion.
	1.2.4	Respiratory values and capacities: tidal volume, inspiratory reserve volume, expiratory reserve volume, residual volume, vital capacity, inspiratory capacity, functional residual capacity, total lung capacity.
	1.2.5	The anatomical components and structure of the cardio vascular system to include, the heart – atria, ventricles, valves, septum, atrioventricular (AV) and sinoatrial (SA) nodes, myocardia – blood, and blood vessels (arteries, veins, and capillaries).

Subject content	What students need to learn
	1.2.6 The physiology of the cardiovascular system with regards to the cardiac cycle, systemic and pulmonary circulation, venous return, vascular shunting, heart rates, (resting, working, maximum, heart rate reserve and recovery), stroke volume, cardiac output, end diastolic and end systolic volumes.
	1.2.7 Understanding of bradycardia, why it may be beneficial and how, anatomically and physiologically, it may occur.
	1.2.8 The cardiorespiratory and cardiovascular systems and how they respond acutely, both structurally and functionally, to the stress of warming up and immediate physical or sporting activity.
	1.2.9 Understanding of what constitutes an unhealthy lifestyle and its effects on the cardiovascular and cardiorespiratory systems.